

On skewed Gaussian graphical models

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ABSTRACT

We introduce a skewed Gaussian graphical model as an extension to the Gaussian graphical model. One of the appealing properties of the Gaussian distribution is that conditional independence can be fully characterized by the sparseness in the precision matrix. The skewed Gaussian distribution adds a shape parameter to the Gaussian distribution to take into account possible skewness in the data; thus it is more flexible than the Gaussian model. Nevertheless, the appealing property of the Gaussian distribution is retained to a large degree: the conditional independence is still characterized by the sparseness in the parameters, which now include a shape parameter in addition to the precision matrix. As a result, the skewed Gaussian graphical model can be efficiently estimated through a penalized likelihood method just as the Gaussian graphical model. We develop an algorithm to maximize the penalized likelihood based on the alternating direction method of multipliers, and establish the asymptotic normality and variable selection consistency for the new estimator. Through simulations, we demonstrate that our method performs better than the Gaussian and Gaussian copula methods when these distributional assumptions are not satisfied. The method is applied to a breast cancer MicroRNA dataset to construct a gene network, which shows better interpretability than the Gaussian graphical model.

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1. Introduction

Graphical models are a class of statistical models defined by conditional independence between variables which can be represented by graphs. Among graphical models, the Gaussian graphical model has been one of the most popular in many applied fields, including network analysis [40], image processing [31] and statistical machine learning [22]. Although the Gaussian assumption brings great simplicity to the graphical model, it is severely restrictive for many applied problems. As a result, estimation based on the Gaussian graphical model may be biased. This motivates us to consider a model which assumes that the variables follow a multivariate skewed Gaussian distribution. The skewed Gaussian distribution extends the Gaussian distribution by introducing a parameter that characterizes the direction of skewness and a location parameter. Skewed Gaussian distributions give our model more flexibility while retaining similar mathematical simplicity of the Gaussian distribution.

Let $X = (X^1, \dots, X^p)^T$ be a p -dimensional random vector. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph, where $\mathcal{V} = \{1, \dots, p\}$ represents the set of p nodes and $\mathcal{E} = \{(i, j) : 1 \leq i < j \leq p\}$ the set of edges describing the relationship among $X^1, \dots, X^p$.

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The edge between nodes i and j is absent if and only if X^i and X^j are conditional independent given all other components of X . In symbols,

$$(i, j) \notin \mathcal{E} \Leftrightarrow X^i \perp\!\!\!\perp X^j \mid X^{-(i,j)},$$

where $X^{-(i,j)}$ represents vector X with two elements X^i and X^j removed.

Under the Gaussian graphical model, conditional independence is determined solely by the zero entries of the precision matrix. Specifically, if X follows a multivariate Gaussian distribution with covariance matrix Σ , and if $\Theta = \Sigma^{-1}$ is the precision matrix with θ_{ij} as its (i, j) th entry, then

$$X^i \perp\!\!\!\perp X^j \mid X^{-(i,j)} \Leftrightarrow \theta_{ij} = 0.$$

Thus, under the Gaussian distribution assumption, the estimation of $\mathcal{G}$ is equivalent to identifying the zero entries in the precision matrix Θ . There have been many recent developments regarding the efficient estimation of Gaussian graphical model. Penalized maximum likelihood estimation with ℓ_1 -penalty is considered in [42]. A neighborhood selection scheme using lasso [33] is proposed in [30]. A block interior point procedure is used to maximize the ℓ_1 -penalized log-likelihood in [6]. A fast blockwise coordinate descent algorithm called graphical lasso is developed for solving the ℓ_1 -penalized log-likelihood maximization problem efficiently in [14]. A constrained ℓ_1 -minimization method for inverse covariance matrix estimation is introduced in [10]. Besides ℓ_1 -penalty, sparsity can also be introduced by other means. For example, the hard thresholding technique is employed in [7,8]; a Dantzig selector [11] type modification is used to replace the lasso selection by in [41].

Since the Gaussian assumption is restrictive in many applications, various adaptations have been taken to extend the Gaussian graphical model. One of the solutions is to assume a Gaussian copula model, which allows data to be non-Gaussian but requires it to be multivariate Gaussian after marginal monotone transformation, see [19,27,28,37]. Other works on non-Gaussian graphical models include [1,13,25,26,34,38]. Although the skewed Gaussian distribution has been proposed by [2] more than three decades ago and its advantages have been well documented [3,5,20], this flexible framework has never been adopted in statistical graphical models. Since the skewed Gaussian distribution enjoys several statistical properties parallel to the Gaussian distribution, it is a convenient joint distribution to relax the Gaussian assumption in the Gaussian graphical model. Like the Gaussian graphical models, in the skewed Gaussian graphical models conditional independence is also formulated by a sparse parameter space. This reduces the difficult task of characterizing conditional independence to parametric sparse estimation. Compared with the Gaussian copula graphical model, the skewed Gaussian graphical model allows skewness to occur along an arbitrary direction in the X -space, whereas the former only allows skewness to occur in the coordinates of X . Even with this flexibility, the skewed Gaussian graphical model is still a parametric model, which is more parsimonious than the semiparametric copula graphical model.

The idea of our method is outlined as follows. Compared to a Gaussian graphical model, in a skewed Gaussian graphical model there is an additional shape parameter when evaluating conditional independence. Nevertheless, there is still a simple relation between the sparse parameter space and conditional independence: in this case conditional independence is determined by the sparseness in both the precision matrix and the shape parameter. We introduce a penalized likelihood to induce sparsity in both parameters and design its structure in such a way that the precision matrix and the shape parameter are separated, which allows us to efficiently perform the penalized optimization by the alternating direction method of multipliers [17].

2. Skewed Gaussian distribution and conditional independence

We first give the definition of the multivariate skewed Gaussian distribution introduced by [2], which is also called skew-normal distribution. In the following, we use a superscript to indicate the components of a vector, and a subscript to index subjects in a sample. For example, if $X_1, \dots, X_n$ are a sample of vectors, then X_i^j denotes the i th component of X_i . If X is a single vector, then X^i denotes the i th component of X . The symbol $\phi_p(x, \Sigma)$ stands for the p -dimensional normal density with zero mean and covariance matrix Σ and $\Phi(\cdot)$ the cumulative distribution of $N(0, 1)$. Let $\sigma_{11}, \dots, \sigma_{pp}$ denote the diagonal elements of Σ , and let $\text{diag}(\sigma_{11}^{-1/2}, \dots, \sigma_{pp}^{-1/2})$ denote the diagonal matrix with $\sigma_{11}^{-1/2}, \dots, \sigma_{pp}^{-1/2}$ as its diagonal elements.

Definition 1. If a p -dimensional random vector $X = (X^1, \dots, X^p)^\top$ has the density

$$p(x, \xi, \Sigma, \alpha) = 2\phi_p(x - \xi, \Sigma)\Phi(\alpha^\top \text{diag}(\sigma_{11}^{-1/2}, \dots, \sigma_{pp}^{-1/2})(x - \xi)), \quad (1)$$

where α is a p -dimensional vector, then X is said to have a multivariate skewed Gaussian distribution, denoted by $SN_p(\xi, \Sigma, \alpha)$. For simplicity, Σ is assumed to be nonsingular.

In this definition, the vector α represents the direction of skewness and is referred to as the shape parameter. When $\alpha = 0$, density (1) reduces to $N(\xi, \Sigma)$. The parameter Σ in the skewed Gaussian distribution is no longer the covariance matrix of X . We call it the scale matrix. For convenience, we still refer to the matrix $\Sigma^{-1} := \Theta$ as the precision matrix,

bearing in mind that it is no longer the inverse of the covariance matrix of X . Next, we present a proposition which shows that $\alpha^\top X$ is the 1-dimensional projection of X with the highest skewness.

Proposition 1. *If $X \sim SN_p(\xi, \Omega, \alpha)$, then the skewness of $c^\top X$ is maximized by any vector c proportional to α .*

Proof. By Proposition 5 and its extension in [4], if $X \sim SN_p(\xi, \Omega, \alpha)$, then $Y = c^\top X \sim SN_1(\xi_Y, \Omega_Y, \alpha_Y)$, where,

$$\xi_Y = c^\top \xi, \quad \Omega_Y = c^\top \Omega c, \quad \alpha_Y = \frac{(c^\top \Omega c)^{-1/2} (\Omega c)^\top \alpha}{\{1 + \alpha^\top [\Omega - \Omega c (c^\top \Omega c)^{-1} (\Omega c)^\top] \alpha\}^{1/2}}.$$

Moreover, the skewness of the distribution $SN_1(\xi_Y, \Omega_Y, \alpha_Y)$ is

$$\kappa_3 = \frac{4 - \pi}{2} \frac{(\delta \sqrt{2/\pi})^3}{(1 - 2\delta^2/\pi)^{3/2}}, \quad \delta = \frac{\alpha_Y}{\sqrt{1 + \alpha_Y^2}}.$$

Let

$$f_1(\delta) = \frac{4 - \pi}{2} \frac{(\delta \sqrt{2/\pi})^3}{(1 - 2\delta^2/\pi)^{3/2}}, \quad f_2(\alpha_Y) = \frac{\alpha_Y}{\sqrt{1 + \alpha_Y^2}}, \quad f_3(t) = \frac{t}{\{1 + \alpha^\top \Omega \alpha - t^2\}^{1/2}}.$$

Then the skewness κ_3 can be expressed as a function of c as follows

$$\kappa_3(c) = (f_1 \circ f_2 \circ f_3)((c^\top \Omega c)^{-1/2} (\Omega c)^\top \alpha).$$

By simple calculation, f_1, f_2 , and f_3 all have positive derivatives. So $\kappa_3(c)$ is maximized at

$$\max\{(c^\top \Omega c)^{-1/2} (\Omega c)^\top \alpha : c \in \mathbb{R}^p\}.$$

By the Cauchy–Schwarz inequality, this maximum is reached by $c = \alpha$. Note that α_Y is invariant with respect to the scale of c ; that is, $\alpha_Y(c) = \alpha_Y(sc)$ for any $s > 0$. Hence κ_3 is also invariant with respect to the scale of c . So $\kappa_3(c)$ is maximized at any c proportional to α . $\square$

To build the graph, we need to characterize the conditional independence between each pair of random variables given the rest of the random variables. One way of building a graph is to characterize the conditional independence between each pair of random variables given the rest of the random variables. In the Gaussian graphical model, conditional independence between any two nodes given the other nodes can be uniquely characterized by the zero entries in the precision matrix. Therefore, building a graph is transformed into sparse estimation of the precision matrix. The correspondence between conditional independence and zero-entries of the precision matrix is what makes the Gaussian graphical model attractive. In the same vein, the skewed Gaussian distribution allows us to identify conditional independence by the sparseness of the parameter, which in this case, involves both Θ and α . Specifically, as shown in [12], the skewed Gaussian distribution implies the following equivalence. Let $X^{-(i,j)}$ represent the subvector of X with its (i, j) th component removed. Let θ_{ij} be the (i, j) th component of Θ and α_i the i th component of α .

Proposition 2 ([12]). *Suppose $X \sim SN_p(\xi, \Sigma, \alpha)$. Then $X^i \perp\!\!\!\perp X^j \mid X^{-(i,j)}$ if and only if the following two conditions hold simultaneously,*

- (i) $\theta_{ij} = 0$;
- (ii) $\alpha_i \alpha_j = 0$.

The above result yields a simple criterion to construct graphs with the skewed Gaussian variates,

$$(i, j) \in \mathcal{E} \Leftrightarrow \theta_{ij} \neq 0 \text{ or } \alpha_i \alpha_j \neq 0.$$

When the shape parameter α is zero, X is normally distributed, and the above condition reduces to $(i, j) \in \mathcal{E} \Leftrightarrow \theta_{ij} \neq 0$. This coincides with the Gaussian graphical model: if the (i, j) th entry of the precision matrix is 0, the two corresponding variables X^i and X^j are conditionally independent given the variables on other nodes. Hence, the Gaussian graphical model is a special case of the Skewed Gaussian graphical model. The latter model allows more flexibility and should be able to fit a broader class of data than the Gaussian graphical model.

3. Penalized likelihood for the skewed Gaussian graphical model

As suggested by the relations in Proposition 2, we perform sparse estimation of the precision matrix $\Theta = \Sigma^{-1}$ and the shape parameter α to estimate edge set $\mathcal{E}$. The complement of the edge set is

$$\mathcal{E}^c = \{(i, j) \in \mathcal{V} \times \mathcal{V} : X^i \perp\!\!\!\perp X^j \mid X^{-(i,j)}, i < j\}.$$

The log-likelihood function for ξ , Θ and α based on a random sample $X_1, \dots, X_n$ of $X \sim SN_p(\xi, \Sigma, \alpha)$ is

$$\begin{aligned} L(\Theta, \alpha, \xi) &= \sum_{l=1}^n \log(p(X_l, \xi, \Sigma)) \\ &\propto \frac{n}{2} \log |\Theta| - \frac{1}{2} \sum_{l=1}^n (X_l - \xi)^\top \Theta (X_l - \xi) + \sum_{l=1}^n \log[\Phi\{\alpha^\top D(X_l - \xi)\}], \end{aligned}$$

where $D = \text{diag}(\sigma_{11}^{-1/2}, \dots, \sigma_{pp}^{-1/2})$ and $|\cdot|$ represents the determinant of a matrix. To simplify the log-likelihood function, we reparameterize α as $\beta = D\alpha$, and thereby represent the objective function as:

$$\begin{aligned} L(\Theta, \beta, \xi) &\propto \frac{1}{2} \log |\Theta| - \frac{1}{2n} \sum_{l=1}^n (X_l - \xi)^\top \Theta (X_l - \xi) + \frac{1}{n} \sum_{l=1}^n \log[\Phi(\beta^\top (X_l - \xi))] \\ &= \frac{1}{2} \log |\Theta| - \frac{1}{2} \text{tr}\{S(\xi) \Theta\} + \frac{1}{n} \sum_{l=1}^n \log[\Phi(\beta^\top (X_l - \xi))], \end{aligned}$$

where $S(\xi) = 1/n \sum_{l=1}^n (X_l - \xi)(X_l - \xi)^\top$. By construction, $\alpha_i = 0$ if and only if $\beta_i = 0$.

Because we are interested in whether there is an edge between any two nodes in the graph, by [Proposition 2](#), the problem is transformed to the identification positions of the zero entries of Θ and β . In other words, the conditional independence $X^i \perp\!\!\!\perp X^j \mid X^{-(i,j)}$ is determined by the sparseness patterns in both Θ and β . Imposing the ℓ_1 -penalty helps us to uncover this sparse structure

$$\frac{1}{2} \log |\Theta| - \frac{1}{2} \text{tr}\{S(\xi) \Theta\} + \frac{1}{n} \sum_{l=1}^n \log[\Phi(\beta^\top (X_l - \xi))] - \lambda_{1,n} \|\Theta\|_1 - \lambda_{2,n} \|\beta\|_1,$$

where $\lambda_{1,n}, \lambda_{2,n} > 0$ are tuning parameters for Θ and β respectively. To simplify computation, we substitute ξ by its maximum likelihood estimate $\hat{\xi}$, a detailed description of the estimation of the location parameter ξ can be found in the Supplementary Material. This results in the following objective function to minimize:

$$-\log |\Theta| + \text{tr}(S(\hat{\xi}) \Theta) - 2 \sum_{l=1}^n \log[\Phi\{\beta^\top (X_l - \hat{\xi})\}]/n + \lambda_{1,n} \|\Theta\|_1 + \lambda_{2,n} \|\beta\|_1. \quad (2)$$

In this expression, Θ and β appear in separate functions, which greatly simplifies the optimization problem. Once $\hat{\Theta}$ and $\hat{\beta}$ are obtained, we use the set

$$\hat{\mathcal{E}} = \{(i, j) : \hat{\theta}_{ij} \neq 0 \text{ or } \hat{\beta}_i \hat{\beta}_j \neq 0, i < j\}$$

to estimate the true edge set $\mathcal{E}$.

4. Computation

4.1. Alternating direction method of multipliers

We employ the alternating direction method of multipliers (ADMM) to minimize the penalized negative likelihood. The chief motivation for using the ADMM algorithm is to handle the non-smoothness of the objective function introduced by the ℓ_1 -penalties. This algorithm was originally proposed by [\[17\]](#) and has become a popular tool for solving the following convex minimization problem with two blocks of functions and variables:

$$\begin{aligned} &\text{Minimize } f(U) + g(V) \\ &\text{subject to } AU + BV = c, \end{aligned}$$

where f and g are convex functions. We form the augmented Lagrangian,

$$\mathcal{L}_\rho(U, V, W) = f(U) + g(V) + \rho W^\top (AU + BV - c) + (\rho/2) \|AU + BV - c\|_F^2,$$

where $\|\cdot\|_F$ is the Frobenius norm, which is the square root of the sum of the squares of the entries.

The alternating direction method of multipliers algorithm consists of the following iterations,

$$\begin{aligned} U^{k+1} &:= \underset{U}{\operatorname{argmin}} \{f(U) + (\rho/2) \|AU + BV^k - c + W^k\|_F^2\}, \\ V^{k+1} &:= \underset{V}{\operatorname{argmin}} \{g(V) + (\rho/2) \|AU^{k+1} + BV - c + W^k\|_F^2\}, \\ W^{k+1} &:= W^k + AU^{k+1} + BV^{k+1} - c. \end{aligned}$$

4.2. Sparse estimation

Since minimization of the objective function in (2) can be decomposed into the sum of two parts, with the first part depending only on Θ , and the second part depending only on β , the minimizer $(\hat{\Theta}, \hat{\beta})$ of $L_n(\Theta, \beta)$ can be found as follows,

$$\hat{\Theta} = \underset{\Theta=\Theta^\top}{\operatorname{argmin}} \{-\log|\Theta| + \operatorname{tr}(S(\hat{\xi})\Theta) + \lambda_{1,n}\|\Theta\|_1\}, \quad (3)$$

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \{-2n^{-1} \sum_{l=1}^n \log[\Phi\{\beta^\top(X_l - \hat{\xi})\}] + \lambda_{2,n}\|\beta\|_1\}. \quad (4)$$

We now solve these two problems separately. Problem (3) has a very similar form as the optimization problem for the graphical lasso, except that, in the latter, the sample covariance matrix takes the place of $S(\hat{\xi})$. Methods for solving problem (3) have been explored in the literature, for example by GLASSO [14,15] and ADMM algorithm [9]. Here, we used ADMM algorithm to solve it. Problem (4) is also solved by ADMM. Let

$$\begin{aligned} f(U) &= -2/n \sum_{l=1}^n \log[\Phi\{U^\top(X_l - \hat{\xi})\}], \\ g(V) &= \lambda_{2,n}\|V\|_1, \\ \mathcal{L}_{\rho_2}(U, V, W) &= f(U) + g(V) + \rho_2 W^\top(U - V) + (\rho_2/2)\|U - V\|_F^2. \end{aligned}$$

Then the algorithm for β proceeds as follows. We initialize U as p -dimensional vector with each entry generated from uniform distribution between 0 and 1 and initialize V and W as p -dimensional zero vectors respectively. At the k th iteration, we perform the following steps:

1. Update U^k to

$$U^{k+1} = \underset{U}{\operatorname{argmin}} \{-2/n \sum_{l=1}^n \log[\Phi\{U^\top(X_l - \hat{\xi})\}] + (\rho_2/2)\|U - V^k + W^k\|_F^2\};$$

2. For $i \in \{1, \dots, p\}$, update V_i^k to $V_i^{k+1} := S_{\lambda_{2,n}/\rho_2}(U_i^{k+1} + W_i^k)$;
3. Update W^k to $W^{k+1} := W^k + U^{k+1} - V^{k+1}$.

The minimization problem in Step 1 can be solved by the gradient descent algorithm. Repeat Steps 1, 2 and 3 until convergence to obtain the final values of U , V and W , denoted by U^* , V^* and W^* . We take $\hat{\beta} = V^*$ as the sparse estimation of β .

4.3. Scale-invariant refinement of the penalty for β

Since only the direction of β , and more specifically the positions of the zero entries of β , affects the graph, in this subsection we introduce a scale-invariant refinement of the penalty on β – the ratio of the ℓ_1 - and ℓ_2 -norms of β . This ratio has been used as a nonconvex sparsity-inducing penalty in blind source separation problems: see, for example, [39]. With this penalty our objective function becomes:

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \{-2n^{-1} \sum_{l=1}^n \log[\Phi\{\beta^\top(X_l - \hat{\xi})\}] + \lambda_{2,n}\|\beta\|_1/\|\beta\|_F\}. \quad (5)$$

Instead of solving (5) exactly, we propose the following heuristic algorithm to approximate the solution of (5) by taking iterative projections. We refer to the estimator derived by this algorithm as the refined estimator of the skewed Gaussian graphical models.

We initialize U as a p -dimensional random vector with each entry being generated from the uniform distribution between 0 and 1, initialize V as the same vector as U , and initialize W as the p -dimensional zero vector. At the k th iteration, we perform the following operations:

1. Compute $U' = \underset{U}{\operatorname{argmin}} \{-2/n \sum_{l=1}^n \log[\Phi\{U^\top(X_l - \hat{\xi})\}] + (\rho_2/2)\|U - V^k + W^k\|_F^2\}$;
2. Update U^k to $U^{k+1} = c \cdot U'/\|U'\|_F$, where c is a constant related to the ℓ_2 norm of the parameter, which can be chosen in a reasonable range;
3. For $i \in \{1, \dots, p\}$, update V_i^k to $V_i^{k+1} := S_{\lambda_{2,n}/\rho_2}(U_i^{k+1} + W_i^k)$;
4. Update W^k to $W^{k+1} := W^k + U^{k+1} - V^{k+1}$.

The minimization problem in Step 1 can be solved by the gradient descent algorithm. Repeat Steps 1 to 4 until convergence to obtain the final values of U , V and W , denoted by U^* , V^* and W^* . We take $\hat{\beta} = V^*$ as the sparse estimation of β . In principle, we should then optimize over c , but since our experiences indicate that the choice of c does not affect the result in any significant way, we choose $c = 3$ in all our simulation studies.

4.4. Update of location parameter and selection of tuning parameter

In this section, we discuss two more issues in the implementation: the updated estimation of the location parameter and the choice of tuning parameter. Recall that, our estimation procedure has been to first estimate ξ by jointly maximizing the likelihood, and then sparsely estimate β and Θ by maximizing their respective penalized likelihoods. To further improve accuracy, we propose to update $\hat{\xi}$ by re-maximizing the likelihood at Θ and β fixed at their sparse estimates. Finally, we re-perform sparse estimation of Θ and β with ξ fixed at the updated estimate. We refer to the estimator derived with the above procedure as the two-step estimator of the skewed Gaussian graphical model. Thus, based on the choice of penalty, we have the two-step original estimator and the two-step refined estimator.

For the choice of the tuning parameter, we use the modified Bayesian information criterion proposed by [36] to select the tuning parameter for β , and the classic Bayesian information criterion to select the tuning parameter for Θ . Again because the objective function in (2) can be separated into two parts with respect to Θ and β respectively, regularization parameters $\lambda_{1,n}$ and $\lambda_{2,n}$ can be selected with the Bayesian information criterion in a parallel way. Let $\hat{\Theta}(\lambda_{1,n}) = \{\hat{\theta}_{ij}(\lambda_{1,n}) : (i, j) \in \mathcal{V} \times \mathcal{V}\}$ and $\hat{\beta}(\lambda_{2,n}) = \{\hat{\beta}_j : j \in 1, \dots, p\}$ be the estimates of Θ and β using the penalized likelihood method with the choices of certain $\lambda_{1,n}$ and $\lambda_{2,n}$. The classic Bayesian information criterion (BIC) is defined as

$$\text{BIC}(\lambda_{1,n}) = -\log |\hat{\Theta}(\lambda_{1,n})| + \text{tr}(\hat{\Theta}(\lambda_{1,n})S(\hat{\xi})) + \sum_{i,j} I(\hat{\theta}_{ij}(\lambda_{1,n}) \neq 0)(\log(n)/n).$$

See [16], and the modified Bayesian information criterion (mBIC) is defined as

$$\text{mBIC}(\lambda_{2,n}) = -2 \sum_{l=1}^n \log[\Phi\{\hat{\beta}(\lambda_{2,n})^\top (X_l - \hat{\xi})\}]/n + \sum_i I(\hat{\beta}_i(\lambda_{2,n}) \neq 0)(\log(n)/n) \log(\log(p)). \quad (6)$$

The tuning parameter is selected as

$$\hat{\lambda}_{\text{BIC}} = \left(\underset{\lambda_{1,n} \in (0, \infty)}{\text{argmin}} \text{BIC}(\lambda_{1,n}), \underset{\lambda_{2,n} \in (0, \infty)}{\text{argmin}} \text{mBIC}(\lambda_{2,n}) \right).$$

4.5. When to use the skewed Gaussian graphical model

An important practical question is how to choose between conventional Gaussian graphical models and skewed Gaussian graphical models. The situation where the skewed Gaussian graphical model is useful is when the multivariate data is skewed in a certain direction, particularly if that direction is not one of the marginals, which can be corrected by a Gaussian copula model. One way to check this is to generate a sample of random directions, say $v_1, \dots, v_m$, and compute the sample skewness of $v_1^\top X, \dots, v_m^\top X$. If the maximum skewness is large and the maximum direction is not close to one of marginal directions $e_1, \dots, e_p$, then we should use a skewed Gaussian graphical model.

5. Asymptotic theory

5.1. Asymptotic normality

We now derive the asymptotic properties of the proposed estimators, focusing on those of the original estimator. For simplicity, we assume that p is fixed as the sample size n goes to infinity. As mentioned earlier, the objective function (2) can be separated into two parts, (3) and (4), which are treated separately by Theorems 1 and 2 in this section.

In the following, $\text{sign}(t)$ is the subgradient of the absolute value function $|t|$ for $t \in \mathbb{R}$; that is, $\text{sign}(t) = 1$ if $t > 0$, $\text{sign}(t) = -1$ if $t < -1$, and $\text{sign}(t)$ is anywhere between -1 and 1 if $t = 0$. Also, for a matrix A , $\text{vec}(A)$ denotes the vector obtained by stacking up the columns of A ; for a symmetric matrix A , $\text{vech}(A)$ denotes the vector obtained by stacking only the lower triangular part of A . We first state a lemma, which gives the explicit formula for the scale matrix Σ in the skewed Gaussian distribution. Interestingly, to our knowledge, this useful result has never been recorded in the literature.

Lemma 1. If $X \sim SN_p(\xi, \Sigma, \alpha)$, then $\Sigma = E_X\{(X - \xi)(X - \xi)^\top\}$.

Proof. Let μ denote the population mean for $X \sim SN_p(\xi, \Sigma, \alpha)$. Using the cumulant generating function of X given by [4], it is easy to derive the first two cumulants of X as follows:

$$\mu = \xi - (2/\pi)^{1/2} \delta, \quad \text{var}(X) = \Sigma - (2\pi) \delta \delta^\top \quad (7)$$

where $\delta = (1 + \beta^\top \Sigma \beta)^{-1/2} \Sigma \beta$. Note that

$$\begin{aligned} E_X\{(X - \xi)(X - \xi)^\top\} &= E_X\{(X - \mu + \mu - \xi)(X - \mu + \mu - \xi)^\top\} = E_X\{(X - \mu)(X - \mu)^\top\} + (\mu - \xi)(\mu - \xi)^\top \\ &= \text{var}(X) + \{(2/\pi)^{1/2} \delta\} \{(2/\pi)^{1/2} \delta\}^\top, \end{aligned}$$

where the last equality follows from the first equation in (7). By the second equation in (7), the right-hand side above reduces to

$$\Sigma - (2/\pi)\delta\delta^\top + \{(2/\pi)^{1/2}\delta\}\{(2/\pi)^{1/2}\delta\}^\top = \Sigma,$$

as desired. $\square$

Also, since ξ is not necessarily the mean of X , Σ is not necessarily the variance matrix of X . Before proceeding further, we need to introduce the concepts of efficient score and efficient information matrix for a parameter of interest in the presence of a nuisance parameter. Let γ denote $(\text{vech}(\Theta)^\top, \beta^\top)^\top$ and let κ denote $(\xi^\top, \gamma^\top)^\top$. The parameter γ is of dimension $p(p+1)/2 + p$ and the parameter κ is of dimension $p(p+1)/2 + 2p$. Let $s_\xi(\kappa, X) = \partial \log p(X, \kappa)/\partial \xi$ be the score function for ξ and $s_\gamma(\kappa, X) = \partial \log p(X, \kappa)/\partial \gamma$ the score function for γ . The Fisher information matrix for κ can then be written as

$$I(\kappa) = \begin{pmatrix} E(s_\xi(\kappa, X)s_\xi^\top(\kappa, X)), E(s_\xi(\kappa, X)s_\gamma^\top(\kappa, X)) \\ E(s_\gamma(\kappa, X)s_\xi^\top(\kappa, X)), E(s_\gamma(\kappa, X)s_\gamma^\top(\kappa, X)) \end{pmatrix} \equiv \begin{pmatrix} I_{\xi\xi} & I_{\xi\gamma} \\ I_{\gamma\xi} & I_{\gamma\gamma} \end{pmatrix}.$$

We then define the efficient score for ξ in the presence of γ as $s_{\xi\cdot\gamma}(\kappa, X) = s_\xi(\kappa, X) - I_{\xi\gamma}I_{\gamma\gamma}^{-1}s_\gamma(\kappa, X)$, and define the efficient information for ξ in the presence of γ as $I_{\xi\cdot\gamma}(\kappa) = I_{\xi\xi} - I_{\xi\gamma}I_{\gamma\gamma}^{-1}I_{\gamma\xi}$. See, for example, [24], page 281.

We now derive the asymptotic distribution of $\sqrt{n}(\hat{\Theta} - \Theta)$. Let (W, H) be a pair of random matrices whose distribution is specified by $(\text{vec}(W)^\top, \text{vec}(H)^\top)^\top \sim N(0, \Gamma)$, where

$$\Gamma = \begin{pmatrix} \Gamma_{11} & \Gamma_{12} \\ \Gamma_{21} & \Gamma_{22} \end{pmatrix}$$

and

$$\begin{aligned} \Gamma_{11} &= \text{var}\{(X - \xi) \otimes (X - \xi)\}, \quad \Gamma_{12} = \Gamma_{21}^\top = \text{cov}\{(X - \xi) \otimes (X - \xi), (2/\pi)^{1/2}\delta \otimes I_{\xi\cdot\gamma}^{-1}(\kappa)s_{\xi\cdot\gamma}(\kappa, X)\}, \\ \Gamma_{22} &= \text{var}\{(2/\pi)^{1/2}\delta \otimes I_{\xi\cdot\gamma}^{-1}(\kappa)s_{\xi\cdot\gamma}(\kappa, X)\}. \end{aligned}$$

In the above, $\otimes$ represents the Kronecker product, and $\delta = (1 + \beta^\top \Sigma \beta)^{-1/2} \Sigma \beta$.

Theorem 1. If $\sqrt{n}\lambda_{1,n} \rightarrow \lambda_1 > 0$ as $n \rightarrow \infty$, then the estimator obtained by the original method $\hat{\Theta}$ for Θ satisfies

$$\sqrt{n}(\hat{\Theta} - \Theta) \rightarrow \underset{U=U^\top \in \mathbb{R}^{p \times p}}{\text{argmin}} V(U, W, H), \quad (8)$$

where $\rightarrow$ represents convergence in distribution and

$$\begin{aligned} V(U, W, H) &= \text{tr}(U\Sigma U\Sigma)/2 + \text{tr}(UW) + \text{tr}(UH^\top) + \text{tr}(UH) \\ &\quad + \lambda_1 \sum_{i=1}^p \sum_{j=1}^p \{u_{ij} \text{sign}(\theta_{ij}) I(\theta_{ij} \neq 0) + |u_{ij}| I(\theta_{ij} = 0)\}, \end{aligned} \quad (9)$$

in which u_{ij} is the (i, j) th entry of $p \times p$ symmetric matrix U .

Proof. For any symmetric matrix $U \in \mathbb{R}^{p \times p}$, let

$$\begin{aligned} V_n(U) &= [-\log |\Theta + n^{-1/2}U| + \text{tr}\{(\Theta + n^{-1/2}U)S(\hat{\xi})\} \\ &\quad + \lambda_{1,n} \sum_{i=1}^p \sum_{j=1}^p |\theta_{ij} + n^{-1/2}u_{ij}|] - [-\log |\Theta| + \text{tr}\{\Theta S(\hat{\xi})\}] + \lambda_{1,n} \sum_{i=1}^p \sum_{j=1}^p |\theta_{ij}| \\ &= -(\log |\Theta + n^{-1/2}U| - \log |\Theta|) + [\text{tr}\{(\Theta + n^{-1/2}U)S(\hat{\xi})\} - \text{tr}\{\Theta S(\hat{\xi})\}] \\ &\quad + (\lambda_{1,n} \sum_{i=1}^p \sum_{j=1}^p |\theta_{ij} + n^{-1/2}u_{ij}| - \lambda_{1,n} \sum_{i=1}^p \sum_{j=1}^p |\theta_{ij}|) \\ &= \text{-(I)} + \text{(II)} + \text{(III)}. \end{aligned} \quad (10)$$

Note that

$$\log |\Theta + n^{-1/2}U| - \log |\Theta| = \log |I_p + n^{-1/2}\Sigma^{1/2}U\Sigma^{1/2}| = \sum_{i=1}^p \log\{1 + n^{-1/2}\sigma_i(\Sigma^{1/2}U\Sigma^{1/2})\},$$

where I_p is the $p \times p$ identity matrix and $\sigma_i(\cdot)$ is the i th largest eigenvalue of a matrix. Since

$$\log\{1 + n^{-1/2}\sigma_i(\Sigma^{1/2}U\Sigma^{1/2})\} = n^{-1/2}\sigma_i(\Sigma^{1/2}U\Sigma^{1/2}) - n^{-1}\sigma_i^2(\Sigma^{1/2}U\Sigma^{1/2})/2 + o(n^{-1}),$$

we have

$$\begin{aligned} (I) &= \sum_{i=1}^p n^{-1/2} \sigma_i(\Sigma^{1/2} U \Sigma^{1/2}) - \sum_{i=1}^p n^{-1} \sigma_i^2(\Sigma^{1/2} U \Sigma^{1/2})/2 + o(n^{-1}) \\ &= n^{-1/2} \text{tr}(\Sigma^{1/2} U \Sigma^{1/2}) - n^{-1} \text{tr}(\Sigma^{1/2} U \Sigma U \Sigma^{1/2})/2 + o(n^{-1}) = n^{-1/2} \text{tr}(U \Sigma) - n^{-1} \text{tr}(U \Sigma U \Sigma)/2 + o(n^{-1}). \end{aligned} \quad (11)$$

The second term on the right hand side of (10) can be re-expressed as

$$(II) = n^{-1/2} \text{tr}\{U S(\hat{\xi})\} = n^{-1/2} \text{tr}(U \Sigma) + n^{-1/2} \text{tr}[U \{S(\hat{\xi}) - \Sigma\}]. \quad (12)$$

Note that $S(\hat{\xi}) - \Sigma$ can be further decomposed as,

$$\begin{aligned} S(\hat{\xi}) - \Sigma &= n^{-1} \sum_{l=1}^n (X_l - \hat{\xi})(X_l - \hat{\xi})^\top - \Sigma \\ &= n^{-1} \sum_{l=1}^n (X_l - \xi)(X_l - \xi)^\top - \Sigma + n^{-1} \sum_{l=1}^n \{X_l - \xi\}(\xi - \hat{\xi})^\top \\ &\quad + n^{-1} \sum_{l=1}^n (\xi - \hat{\xi})(X_l - \xi)^\top + (\xi - \hat{\xi})(\xi - \hat{\xi})^\top \\ &= n^{-1/2} W_n + n^{-1/2} H_n^\top + n^{-1/2} H_n + O_p(n^{-1}), \end{aligned} \quad (13)$$

where $W_n = n^{1/2} \{n^{-1} \sum_{l=1}^n (X_l - \xi)(X_l - \xi)^\top - \Sigma\}$ and $H_n = n^{-1/2} \sum_{l=1}^n (\xi - \hat{\xi})(X_l - \xi)^\top$.

Let E_n denote the sample mean operator, that is, $E_n(g(X)) = n^{-1} \sum_{l=1}^n g(X_l)$ for a function $g(X)$ of X . By Theorem 9.7 of [24], we have

$$\hat{\xi} = \xi + I_{\xi, \gamma}(\kappa)^{-1} n^{-1} \sum_{l=1}^n s_{\xi, \gamma}(\kappa, X_l) + o_p(n^{-1/2}). \quad (14)$$

Then H_n can be written as

$$\begin{aligned} H_n &= n^{-1/2} \sum_{l=1}^n (\xi - \hat{\xi})(X_l - \xi)^\top = n^{1/2} (\xi - \hat{\xi}) n^{-1} \sum_{l=1}^n (X_l - \xi)^\top \\ &= n^{1/2} \{-I_{\xi, \gamma}(\kappa)^{-1} n^{-1} \sum_{l=1}^n s_{\xi, \gamma}(\kappa, X_l) + o_p(n^{-1/2})\} \{\mu - \xi + o_p(1)\}^\top \\ &= -n^{1/2} E_n\{I_{\xi, \gamma}(\kappa)^{-1} s_{\xi, \gamma}(\kappa, X)\} \{\mu - \xi\}^\top + o_p(1) = n^{1/2} E_n\{I_{\xi, \gamma}(\kappa)^{-1} s_{\xi, \gamma}(\kappa, X)\} (2/\pi)^{1/2} \delta^\top + o_p(1), \end{aligned}$$

where the last equality follows from the first equation in (7). As a result,

$$\begin{pmatrix} W_n \\ H_n \end{pmatrix} = \sqrt{n} \begin{pmatrix} E_n\{(X - \xi)(X - \xi)^\top - \Sigma\} \\ E_n\{I_{\xi, \gamma}(\kappa)^{-1} s_{\xi, \gamma}(\kappa, X)(2/\pi)^{1/2} \delta^\top\} \end{pmatrix} + o_p(1).$$

After vectorizing W_n and H_n , we have

$$\begin{pmatrix} \text{vec}(W_n) \\ \text{vec}(H_n) \end{pmatrix} = \sqrt{n} E_n \left(\begin{pmatrix} (X - \xi) \otimes (X - \xi) - \text{vec}(\Sigma) \\ \{(2/\pi)^{1/2} \delta\} \otimes \{I_{\xi, \gamma}(\kappa)^{-1} s_{\xi, \gamma}(\kappa, X)\} \end{pmatrix} \right) + o_p(1).$$

By Slutsky's theorem,

$$\begin{pmatrix} \text{vec}(W_n) \\ \text{vec}(H_n) \end{pmatrix} \rightarrow N \left(0, \text{var} \left(\begin{pmatrix} (X - \xi) \otimes (X - \xi) - \text{vec}(\Sigma) \\ \{(2/\pi)^{1/2} \delta\} \otimes \{I_{\xi, \gamma}(\kappa)^{-1} s_{\xi, \gamma}(\kappa, X)\} \end{pmatrix} \right) \right) = N(0, \Gamma),$$

where $\rightarrow$ stands for convergence in distribution. So if we let W and H be $p \times p$ random matrices such that $(\text{vec}(W)^\top, \text{vec}(H)^\top)^\top$ has the above multivariate normal distribution, then $n^{1/2}(II)$ converges in distribution to

$$\text{tr}(U \Sigma) + \text{tr}(U(W + H + H^\top)).$$

The third item on the right hand side of (10) can be written in the form

$$(III) = \lambda_{1,n} \sum_{i=1}^p \sum_{j=1}^p (|\theta_{ij} + n^{-1/2} u_{ij}| - |\theta_{ij}|) = n^{-1/2} \lambda_{1,n} \sum_{i=1}^p \sum_{j=1}^p \{u_{ij} \text{sign}(\theta_{ij}) I(\theta_{ij} \neq 0) + |u_{ij}| I(\theta_{ij} = 0)\}. \quad (15)$$

Combining (15) with (11), (12) and (13), we have

$$\begin{aligned} nV_n(U) &= \text{tr}(U\Sigma U\Sigma)/2 + \text{tr}(UW_n) + \text{tr}(UH_n^\top) + \text{tr}(UH_n) \\ &\quad + \sqrt{n}\lambda_{1,n} \sum_{i=1}^p \sum_{j=1}^p \{u_{ij} \text{sign}(\theta_{ij})I(\theta_{ij} \neq 0) + |u_{ij}|I(\theta_{ij} = 0)\} + o_p(1). \end{aligned}$$

Therefore, $nV_n(U)$ converges in distribution to $V(U, W, H)$, where

$$\begin{aligned} V(U, W, H) &= \text{tr}(U\Sigma U\Sigma)/2 + \text{tr}(UW) + \text{tr}(UH^\top) + \text{tr}(UH) \\ &\quad + \lambda_1 \sum_{i=1}^p \sum_{j=1}^p \{u_{ij} \text{sign}(\theta_{ij})I(\theta_{ij} \neq 0) + |u_{ij}|I(\theta_{ij} = 0)\}. \end{aligned}$$

Since both $V(U, W, H)$ and $nV_n(U)$ are convex and $V(U, W, H)$ has a unique minimum, it follows that

$$\underset{U=U^\top}{\text{argmin}} nV_n(U) = \sqrt{n}(\hat{\Theta} - \Theta) \rightarrow \underset{U=U^\top}{\text{argmin}} V(U, W, H),$$

where $\rightarrow$ represents convergence in distribution. $\square$

We observe that the asymptotic distribution of graphical lasso in [42] does not have the second and the third term in (9). The asymptotic distribution $\hat{\beta}$ is covered by the next theorem. The $\sqrt{n}$ -consistency of $\hat{\xi}$ follows from the usual asymptotic theory of maximum likelihood estimation. See, for example, Chapter 8 of [24].

Theorem 2. If $\sqrt{n}\lambda_{2,n} \rightarrow \lambda_2 > 0$ as $n \rightarrow \infty$, then the estimator obtained by the original method $\hat{\beta}$ for β satisfies

$$\sqrt{n}(\hat{\beta} - \beta) \rightarrow \underset{\tilde{U} \in \mathbb{R}^p}{\text{argmin}} \tilde{V}(\tilde{U}, R), \quad (16)$$

where $\rightarrow$ represents convergence in distribution and

$$\tilde{V}(\tilde{U}, R) = 2R^\top \tilde{U} - \tilde{U}^\top E[\partial^2 L\{\beta^\top(X - \xi)\}/\partial\beta\partial\beta^\top]\tilde{U} + \lambda_2 \sum_{j=1}^p \{\tilde{u}_j \text{sign}(\beta_j)I(\beta_j \neq 0) + |\tilde{u}_j|I(\beta_j = 0)\},$$

in which $L(x)$ denotes $\log\{\Phi(x)\}$, R is a multivariate normal random variable with mean 0 and variance $\sigma_R^2 = \text{var}\{\partial L\{\beta^\top(X - \xi)\}/\partial\beta\}$, and $\tilde{U}$ is a p -dimensional vector.

Proof. Let

$$\begin{aligned} \tilde{V}_n(\tilde{U}) &= -2n^{-1} \sum_{l=1}^n \log[\Phi\{(\beta + n^{-1/2}\tilde{U})^\top(X_l - \hat{\xi})\}] + \lambda_{2,n} \sum_{j=1}^p |\beta_j + n^{-1/2}\tilde{u}_j| \\ &\quad - (-2n^{-1} \sum_{l=1}^n \log[\Phi\{\beta^\top(X_l - \hat{\xi})\}] + \lambda_{2,n} \sum_{j=1}^p |\beta_j|) \\ &= -(2n^{-1} \sum_{l=1}^n \log[\Phi\{(\beta + n^{-1/2}\tilde{U})^\top(X_l - \hat{\xi})\}] - 2n^{-1} \sum_{l=1}^n \log[\Phi\{\beta^\top(X_l - \hat{\xi})\}]) \\ &\quad + (\lambda_{2,n} \sum_{j=1}^p |\beta_j + n^{-1/2}\tilde{u}_j| - \lambda_{2,n} \sum_{j=1}^p |\beta_j|) = -(\text{I}) + (\text{II}). \end{aligned} \quad (17)$$

By Taylor's expansion,

$$\begin{aligned} E_n[L\{(\beta + n^{-1/2}\tilde{U})^\top(X - \hat{\xi})\}] &= E_n[L\{\beta^\top(X - \hat{\xi})\}] + n^{-1/2}E_n[\partial L\{\beta^\top(X - \hat{\xi})\}/\partial\beta^\top]\tilde{U} \\ &\quad + (n^{-1}/2)\tilde{U}^\top E_n[\partial^2 L\{\beta^\top(X - \hat{\xi})\}/\partial\beta\partial\beta^\top]\tilde{U} + o(n^{-1}). \end{aligned} \quad (18)$$

Since the sequence of functions $\xi \mapsto E_n[\partial^2 L\{\beta^\top(X - \hat{\xi})\}/\partial\beta\partial\beta^\top]$, $n \in \{1, 2, \dots\}$ is stochastically equicontinuous in a neighborhood of the true value of ξ , by Proposition 8.6 in [24], the third term on the right hand side of (18) can be expressed as

$$(n^{-1}/2)\tilde{U}^\top E_n[\partial^2 L\{\beta^\top(X - \hat{\xi})\}/\partial\beta\partial\beta^\top]\tilde{U} = (n^{-1}/2)\tilde{U}^\top E[\partial^2 L\{\beta^\top(X - \xi)\}/\partial\beta\partial\beta^\top]\tilde{U} + o_p(n^{-1}).$$

By Taylor's mean value theorem, the second term on the right hand side of (18) is

$$\begin{aligned} n^{-1/2}E_n[\partial L\{\beta^\top(X - \hat{\xi})\}/\partial\beta^\top]\tilde{U} &= n^{-1/2}E_n[\partial L\{\beta^\top(X - \xi)\}/\partial\beta^\top]\tilde{U} \\ &\quad + n^{-1/2}(\hat{\xi} - \xi)^\top E_n[\partial^2 L\{\beta^\top(X - \xi^\dagger)\}/\partial\xi\partial\beta^\top]\tilde{U}, \end{aligned}$$

for some $\xi^\dagger$ between $\hat{\xi}$ and ξ . Again, because the sequence of functions $\xi \mapsto E_n[\partial^2 L\{\beta^\top(X - \hat{\xi})\}/\partial \xi \partial \beta^\top]$, $n \in \{1, 2, \dots\}$ is stochastically equicontinuous in a neighborhood of the true value of ξ , we have

$$E_n[\partial^2 L\{\beta^\top(X - \xi^\dagger)\}/\partial \xi \partial \beta^\top] = E_n[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \xi \partial \beta^\top] + o_p(1).$$

Hence the first term on the right hand side of (17) can be expressed as

$$(I) = 2n^{-1/2} E_n[\partial L\{\beta^\top(X - \xi)\}/\partial \beta] \tilde{U} + 2n^{-1/2} (\hat{\xi} - \xi)^\top E_n[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \xi \partial \beta^\top] \tilde{U} + n^{-1} \tilde{U}^\top E_n[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \beta \partial \beta^\top] \tilde{U} + o_p(n^{-1}). \quad (19)$$

The second term on the right-hand side of (17) can be written as

$$(II) = \lambda_{2,n} \sum_{j=1}^p (|\beta_j + n^{-1/2} \tilde{u}_j| - |\beta_j|) = n^{-1/2} \lambda_{2,n} \sum_{j=1}^p \{\tilde{u}_j \text{sign}(\beta_j) I(\beta_j \neq 0) + |\tilde{u}_j| I(\beta_j = 0)\}. \quad (20)$$

Substituting (19) and (20) into (17), we have

$$\begin{aligned} n\tilde{V}_n(\tilde{U}) &= -2n^{1/2} E_n[\partial L\{\beta^\top(X - \xi)\}/\partial \beta^\top] \tilde{U} - 2n^{1/2} (\hat{\xi} - \xi)^\top E_n[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \xi \partial \beta^\top] \tilde{U} \\ &\quad - \tilde{U}^\top E_n[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \beta \partial \beta^\top] \tilde{U} + \sqrt{n} \lambda_{2,n} \sum_{j=1}^p \{\tilde{u}_j \text{sign}(\beta_j) I(\beta_j \neq 0) \\ &\quad + |\tilde{u}_j| I(\beta_j = 0)\} + o_p(1). \end{aligned} \quad (21)$$

By the Central Limit Theorem, the first term on the right-hand side of (21) converges in distribution to $2R^\top \tilde{U}$, where R is a multivariate normal random vector with zero mean and variance $\sigma_R^2 = \text{var}(\partial L\{\beta^\top(X - \xi)\}/\partial \beta)$.

By Slutsky's theorem, the second term in (21) is of the order $o_p(1)$. Therefore $n\tilde{V}_n(\tilde{U}) \rightarrow \tilde{V}(\tilde{U}, R)$ in distribution, where

$$\tilde{V}(\tilde{U}, R) = 2R^\top \tilde{U} - \tilde{U}^\top E[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \beta \partial \beta^\top] \tilde{U} + \lambda_2 \sum_{j=1}^p \{\tilde{u}_j \text{sign}(\beta_j) I(\beta_j \neq 0) + |\tilde{u}_j| I(\beta_j = 0)\}.$$

Since $\sqrt{n}(\hat{\beta} - \beta) \rightarrow \underset{\tilde{U}}{\text{argmin}} n\tilde{V}_n(\tilde{U})$, $n\tilde{V}_n(\tilde{U})$ converges in distribution to $\tilde{V}(\tilde{U}, R)$ and both functions are convex, we have the desired result. $\square$

Corollary 1. With $V(U, W, H)$, $\tilde{V}(\tilde{U}, R)$, and corresponding random vectors W, H and R defined in Theorems 1 and 2, we have

$$\sqrt{n} \begin{pmatrix} \hat{\Theta} - \Theta \\ \hat{\beta} - \beta \end{pmatrix} \rightarrow \begin{pmatrix} \underset{U=U^\top \in \mathbb{R}^{p \times p}}{\text{argmin}} V(U, W, H) \\ \underset{\tilde{U}}{\text{argmin}} \tilde{V}(\tilde{U}, R) \end{pmatrix},$$

where $\rightarrow$ stands for convergence in distribution, and $(\text{vec}(W), \text{vec}(H), R) \sim N(0, \Psi)$,

$$\Psi = \begin{pmatrix} \Gamma_{11} & \Gamma_{12} & \Psi_{13} \\ \Gamma_{21} & \Gamma_{22} & \Psi_{23} \\ \Psi_{31} & \Psi_{32} & \Psi_{33} \end{pmatrix},$$

$$\Psi_{13} = \Psi_{31}^\top = \text{cov}\{(X - \xi) \otimes (X - \xi), \partial L\{\beta^\top(X - \xi)\}/\partial \beta\},$$

$$\Psi_{23} = \Psi_{32}^\top = \text{cov}\{(2/\pi)^{1/2} \delta \otimes I_{\xi, \gamma}^{-1}(\kappa) s_{\xi, \gamma}(\kappa, X), \partial L\{\beta^\top(X - \xi)\}/\partial \beta\},$$

$$\Psi_{33} = \text{var}(\partial L\{\beta^\top(X - \xi)\}/\partial \beta),$$

and $\Gamma_{11}, \Gamma_{12}, \Gamma_{21}$ and Γ_{22} have been specified before.

Using the above results, we can show that there is a positive probability to estimate a parameter as 0 when it is 0, which is called the sparse property.

5.2. Sparsity

In this subsection, we develop the sparse property of our estimates. In [23], it is shown that limiting distributions of the lasso estimates put positive mass at 0 when the true value of the parameter is 0. As shown below, our estimates also enjoy this appealing property. In the following, for a set A , $\text{card}(A)$ stands for the cardinality of A , and $\mathbb{S}^{p \times p}$ represents the set of all $p \times p$ symmetric matrices.

Theorem 3. Suppose $U^0 = \underset{U=U^\top \in \mathbb{R}^{p \times p}}{\text{argmin}} V(U, W, H)$ in (8), which is the limit of $\sqrt{n}(\hat{\Theta} - \Theta)$, and $\tilde{U}^0 = \underset{\tilde{U} \in \mathbb{R}^p}{\text{argmin}} \tilde{V}(\tilde{U}, R)$ in (16), which is the limit of $\sqrt{n}(\hat{\beta} - \beta)$. Let $B_0 = \{(i, j) : \theta_{ij} = 0\}$ and $C_0 = \{i : \beta_i = 0\}$. Then the

following assertions hold:

- (i) $\Pr(U_{ij}^0 = 0, \text{ if } (i, j) \in B_0) > 0$;
- (ii) $\Pr(\tilde{U}_i^0 = 0, \text{ if } i \in C_0) > 0$;
- (iii) For any $\epsilon > 0$, there is a $\lambda_1 > 0$ such that $\Pr(U_{ij}^0 = 0, (i, j) \in B_0, i < j) > 1 - \epsilon$;
- (iv) For any $\epsilon > 0$, there is a $\lambda_2 > 0$ such that $\Pr(\tilde{U}_i^0 = 0, i \in C_0) > 1 - \epsilon$.

Proof. (i) For $f : \mathbb{R}^{p \times p} \rightarrow \mathbb{R}$, let $\partial f / \partial A$ be the $s \times s$ matrix whose (i, j) th entry is $\partial f / \partial A_{ij}$. Let $\partial^L f / \partial A_{ij}$ and $\partial^R f / \partial A_{ij}$ denote the left- and right-partial derivatives with respect to the (i, j) th entry of matrix A . When f is differentiable with respect to A_{ij} , we have $\partial^L f / \partial A_{ij} = \partial^R f / \partial A_{ij} = \partial f / \partial A_{ij}$. Moreover, for a specific value A_{ij}^0 of A_{ij} , let $\partial f(A_{ij}^0) / \partial A_{ij}$ denote the evaluation of $\partial f / \partial A_{ij}$ at A_{ij}^0 . The same applies to $\partial^L f / \partial A_{ij}$ and $\partial^R f / \partial A_{ij}$. Let $\mathbb{S}^{p \times p}(B_0)$ denote the set of matrices in $\mathbb{S}^{p \times p}$ whose (i, j) th entry is 0 whenever $(i, j) \in B_0$. Note that $U_{ij}^0 = 0$ for $(i, j) \in B_0$ if and only if $V(U)$ is minimized within $\mathbb{S}^{p \times p}(B_0)$, which is equivalent to

$$\begin{cases} \partial V(U_{ij}^0) / \partial U_{ij} = 0, & \text{if } (i, j) \in B_0^c, i < j, \\ \partial^L V(U_{ij}^0) / \partial U_{ij} \leq 0 \leq \partial^R V(U_{ij}^0) / \partial U_{ij}, & \text{if } (i, j) \in B_0, i < j, \end{cases} \quad (22)$$

where at least one of the two $\leq$ in the second line is $<$. Let

$$\begin{aligned} J(U) &= \text{tr}(U \Sigma U \Sigma) / 2 + \text{tr}\{U(W + H^\top + H)\}, \\ K(U) &= \lambda_1 \sum_{i=1}^p \sum_{j=1}^p \{u_{ij} \text{sign}(\theta_{ij}) I(\theta_{ij} \neq 0) + |u_{ij}| I(\theta_{ij} = 0)\}. \end{aligned}$$

Then $\partial J(U) / \partial U = \Sigma U \Sigma / 2 + W + H^\top + H$. For $(i, j) \in B_0^c$, $K(U)$ is differentiable with respect to u_{ij} and $\partial_{u_{ij}} K(U) / \partial U = \lambda_1 \text{sign}(\theta_{ij})$. For $(i, j) \in B_0$, $K(U)$ is not differentiable with respect to u_{ij} , but has left and right derivatives, given by $\partial_{u_{ij}}^L K(U) / \partial U = -\lambda_1$ and $\partial_{u_{ij}}^R K(U) / \partial U = \lambda_1$. Consequently, (22) reduces to

$$\begin{cases} (\Sigma U \Sigma)_{ij} / 2 + W_{ij} + H_{ji} + H_{ij} + \lambda_1 \text{sign}(\theta_{ij}) = 0, & \text{if } (i, j) \in B_0^c, i < j, \\ -\lambda_1 \leq (\Sigma U \Sigma)_{ij} / 2 + W_{ij} + H_{ji} + H_{ij} \leq \lambda_1, & \text{if } (i, j) \in B_0, i < j, \end{cases} \quad (23)$$

where $(\cdot)_{ij}$ stands for the (i, j) th entry of a matrix and $U \in \mathbb{S}^{p \times p}(B_0)$. Now consider the event

$$G = \{W_{ij} + H_{ji} + H_{ij} : i < j\} : \text{there exists at least one solution } U \text{ for Eq. (23)}.$$

We need to prove $\Pr(G) > 0$, which is true if G contains a nonempty open set. Since U is symmetric, the number of free parameters in U is the same as the number of equation constraints in the first line of (23). Since Σ is nonsingular, for any $\{W_{ij} + H_{ji} + H_{ij} : (i, j) \in B_0^c, i < j\}$, there exists a unique U that satisfies the first line of (23). In addition, the solution U is a linear function of $\{W_{ij} + H_{ji} + H_{ij} : (i, j) \in B_0^c, i < j\}$. Let U be the solution to the first line of (23). Since the map

$$\tau : \{W_{ij} + H_{ji} + H_{ij} : i < j\} \mapsto \{(\Sigma U \Sigma)_{ij} / 2 + W_{ij} + H_{ji} + H_{ij} : i < j, (i, j) \in B_0\}$$

from $\mathbb{R}^{p(p-1)/2}$ to $\mathbb{R}^{\text{card}(B_0)/2}$ is continuous, the set $\tau^{-1}\{(-\lambda_1, \lambda_1)^{\text{card}(B_0)/2}\}$ is open in $\mathbb{R}^{p(p-1)/2}$. Furthermore, the open set is nonempty since if we choose

$$W_{ij} + H_{ji} + H_{ij} = -(\Sigma U \Sigma)_{ij} / 2, \text{ for } (i, j) \in B_0, i < j,$$

then $\tau(\{W_{ij} + H_{ji} + H_{ij} : i < j\}) = 0$. Because $(\text{vec}(W)^\top, \text{vec}(H)^\top)^\top$ has a multivariate normal distribution with a nonsingular covariance matrix, any nonempty open set in $\mathbb{R}^{p(p-1)/2}$ has positive probability for $W + H^\top + H$.

(ii) For $f : \mathbb{R}^p \rightarrow \mathbb{R}$, let $\partial^L f / \partial t_i$ and $\partial^R f / \partial t_i$ denote the left- and right-partial derivatives with respect to the i th entry of t . Let $\mathbb{R}^p(C_0)$ denote the vectors whose i th entry is zero whenever $i \in C_0$. Note that $\tilde{V}(\cdot)$ is minimized at $\tilde{U} \in \mathbb{R}^p(C_0)$ if and only if

$$\begin{cases} \partial \tilde{V}(\tilde{U}_i) / \partial \tilde{U} = 0, & \text{if } i \in C_0^c, \\ \partial^L \tilde{V}(\tilde{U}_i) / \partial \tilde{U}_i \leq 0 \leq \partial^R \tilde{V}(\tilde{U}_i) / \partial \tilde{U}_i, & \text{if } i \in C_0, \end{cases} \quad (24)$$

where at least one of the two $\leq$ in the second line is $<$.

Let

$$\begin{aligned} M(\tilde{U}) &= 2R^\top \tilde{U} - \tilde{U}^\top E[\partial^2 L(\beta^\top(X - \xi)) / \partial \beta \partial \beta^\top] \tilde{U}, \\ N(\tilde{U}) &= \lambda_2 \sum_{j=1}^p \{\tilde{u}_j \text{sign}(\beta_j) I(\beta_j \neq 0) + |\tilde{u}_j| I(\beta_j = 0)\}. \end{aligned}$$

Then

$$\partial M(\tilde{U}) / \partial \tilde{U} = 2R - 2E[\partial^2 L(\beta^\top(X - \xi)) / \partial \beta \partial \beta^\top] \tilde{U}.$$

For $i \in C_0^c$ such that $\beta_i \neq 0$, $N(\tilde{U})$ is differentiable with respect to $\tilde{U}_i$ and $\partial N(\tilde{U})/\partial \tilde{U}_i = \lambda_2 \text{sign}(\beta_i)$. For $i \in C_0$ such that $\beta_i = 0$, $N(\tilde{U})$ is not differentiable with respect to $\tilde{U}_i$, but has left and right derivatives, which are $\partial^L N(\tilde{U})/\partial \tilde{U}_i = -\lambda_2$ and $\partial^R N(\tilde{U})/\partial \tilde{U}_i = \lambda_2$. Thus (24) reduces to

$$\begin{cases} 2R_i - \left(2E[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \beta \partial \beta^\top] \tilde{U} \right)_i + \lambda_2 \text{sign}(\beta_i) = 0, & \text{if } i \in C_0^c, \\ -\lambda_2 \leq 2R_i - \left(2E[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \beta \partial \beta^\top] \tilde{U} \right)_i \leq \lambda_2, & \text{if } i \in C_0, \end{cases} \quad (25)$$

where $(\cdot)_i$ denotes the i th entry of the vector.

Let

$$\tilde{G} = \{R = (R_1, \dots, R_p)^\top : \text{Eq. (25) has at least one solution } \tilde{U}\}.$$

We now show $\Pr(\tilde{G}) > 0$, which holds if $\tilde{G}$ contains a nonempty open set. Since $E[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \beta \partial \beta^\top]$ is nonsingular, any of its submatrix along the diagonal is nonsingular, which implies the first equation in (25) has a solution. Let $\tilde{U}$ be such a solution. Since the mapping

$$\tilde{\tau} : R \mapsto \left\{ 2R_i - \left(2E[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \beta \partial \beta^\top] \tilde{U} \right)_i : i \in C_0 \right\}$$

from $\mathbb{R}^p$ to $\mathbb{R}^{\text{card}(C_0)}$ is continuous, the set $\tilde{\tau}^{-1}[(-\lambda_2, \lambda_2)^{\text{card}(C_0)}]$ is open in $\mathbb{R}^p$. Furthermore, the open set is nonempty since if we choose

$$R_i = \left(E[\partial^2 L\{\beta^\top(X - \xi)\}/\partial \beta \partial \beta^\top] \tilde{U} \right)_i, \text{ for } i \in C_0$$

then $\tau(\{R_i, i \in C_0\}) = 0$. Because R has a multivariate normal distribution with a non-singular covariance matrix, any nonempty open set in $\mathbb{R}^p$ has positive probability, proving $\Pr(\tilde{G}) > 0$.

(iii) For each $(i, j) \in B_0$, $i < j$, let $M_{ij} = (\Sigma U \Sigma)_{ij}/2 + W_{ij} + H_{ji} + H_{ij}$. Then for any $\epsilon > 0$, there is a $\lambda_1^{\tilde{U}} > 0$ such that $\Pr(M_{ij} \in [-\lambda_1^{\tilde{U}}, \lambda_1^{\tilde{U}}]) > 1 - 2\epsilon/\text{card}(B_0)$. Let $\varsigma_1 = \max\{\lambda_1^{\tilde{U}} : (i, j) \in B_0, i < j\}$. Then $\Pr(M_{ij} \in [-\varsigma_1, \varsigma_1]) > 1 - 2\epsilon/\text{card}(B_0)$. Hence

$$\begin{aligned} \Pr(U_{ij}^0 = 0, (i, j) \in B_0, i < j) &\geq \Pr(M_{ij} \in [-\lambda_1^{\tilde{U}}, \lambda_1^{\tilde{U}}], (i, j) \in B_0, i < j) \\ &\geq 1 - \sum_{(i, j) \in B_0, i < j} \{1 - \Pr(M_{ij} \in [-\varsigma_1, \varsigma_1])\} \\ &\geq 1 - \epsilon. \end{aligned}$$

The proof of statement (iv) is similar and is omitted. $\square$

6. Numerical study

In this section, we compare our methods, the two-step original method and the two-step refined method, with the Gaussian graphical model by [42], and the Gaussian copula graphical model by [27,28]. Here, we use both the two-step original estimator and the two-step refined estimator in all simulation settings. We use the receiver operating characteristic curve (ROC) to evaluate the performance of each estimate.

In the standard setting where a single tuning parameter (say λ) is involved, the ROC curve is defined as follows. Let $\hat{\mathcal{E}}(\lambda)$ be the estimated edge set for a fixed λ , and let $\mathcal{E}$ be the true edge set. The false positive rate and the true positive rate at λ are defined by

$$\text{FP}(\lambda) = \frac{\text{card}(\hat{\mathcal{E}}(\lambda) \cap \mathcal{E}^c)}{\text{card}(\mathcal{E}^c)}, \quad \text{TP}(\lambda) = \frac{\text{card}(\hat{\mathcal{E}}(\lambda) \cap \mathcal{E})}{\text{card}(\mathcal{E})}.$$

The ROC curve is then defined as the curve

$$\{(\text{FP}(\lambda), \text{TP}(\lambda)) : \lambda \in \Lambda\},$$

where Λ is some interval in $\mathbb{R}$. Since we have two tuning parameters $\lambda_{1,n}$ and $\lambda_{2,n}$, the above procedure cannot be directly applied. So we adopt the following strategy: we first use the modified Bayesian information criterion (mBIC) described in Eq. (6) to select $\lambda_{2,n}$ and estimate β . With $\lambda_{2,n}$ and β fixed, we then estimate the precision matrix Θ in our model for a range of $\lambda_{1,n}$ and report the ROC curve as a function of $\lambda_{1,n}$.

Our first model is the skewed Gaussian model:

$$\text{Model I: } X \sim \text{SN}_p(\xi, \Sigma, \alpha).$$

The shape parameter β is generated in two steps. First, set each entry of β as either 0 or 1 according to the Bernoulli distribution with success probability 0.3, and then replace each nonzero entry of β with a number randomly chosen from $\{3, 4, 5\}$. The location parameter ξ is generated from Uniform(0, 10) for lower dimension settings ($p = 50$ and 100). For the settings with $p \geq 200$, we impose sparseness on ξ as follows: we first set each entry of ξ as either 0 or 1 according

to the Bernoulli distribution with success probability 0.1. Next, replace each nonzero entry of ξ with a number randomly chosen from Uniform(3, 5). To accommodate the sparsity in ξ , we update $\hat{\xi}$ by maximizing the penalized likelihood with ℓ_1 penalty and Θ and β fixed at their sparse estimates. The precision matrix is generated as follows. We set

$$\gamma_{ii} = 1, \gamma_{i,i-1} = \gamma_{i-1,i} = 0.4, \gamma_{i,i-2} = \gamma_{i-2,i} = 0.1, \gamma_{i,i-3} = \gamma_{i-3,i} = 0.2,$$

which is simply the inverse covariance matrix for an autoregressive process of order 3. We then set $\theta_{ij} = \gamma_{ij}$ if $\beta_i \beta_j = 0$ and $\theta_{ij} = 0$ if $\beta_i \beta_j \neq 0$.

We consider six scenarios:

$$(p, n) = (50, 1000), (100, 1000), (200, 2000), (300, 2000), (400, 3000), (500, 3000).$$

The averaged ROC curves for the four estimators based on 50 replicates are presented in Fig. 1. It indicates that, in all cases, the skewed Gaussian graphical models and its refinement significantly outperform the Gaussian and the Gaussian copula graphical models. This is because neither the Gaussian nor the Gaussian copula assumption is satisfied by Model I. Moreover, for higher dimensions, the refined version of the skewed Gaussian graphical model performs significantly better than the unrefined version.

Our second model is intended to show whether there is a significant loss of accuracy by our estimators when the Gaussian assumption is satisfied. Consider the model

$$\text{Model II: } X \sim N(\mu, \Theta^{-1}),$$

where the precision matrix Θ is again the inverse covariance matrix of an autoregressive process of order 3, that is,

$$\theta_{ii} = 1, \theta_{i,i-1} = \theta_{i-1,i} = 0.4, \theta_{i,i-2} = \theta_{i-2,i} = \theta_{i,i-3} = \theta_{i-3,i} = 0.2.$$

The parameter μ is generated in the same way as we generated ξ in the Model I. We used the same six (n, p) scenarios as in Model I. The averaged ROC curves based on 50 replicates are presented in Fig. 2.

As shown in 2, under the Gaussian assumption, the four methods all have similar performance. The refined skewed Gaussian graphical model outperforms the Gaussian graphical model and the Gaussian copula graphical model. These three models all outperform the original skewed Gaussian graphical model. However, the difference is not very large, indicating that the loss of accuracy by our original method under the Gaussian assumption is modest.

We next consider a simulation study where the covariance matrix is randomly generated for each simulation. Such a scenario was also used in [10,32]. Consider

$$\text{Model III: } X \sim SN_p(\xi, \Sigma, \alpha),$$

where $\Theta = \Sigma^{-1} = B + \delta I$. Each off-diagonal element in B is generated from a Bernoulli distribution with success probability $p = 0.1$. The diagonal elements of B are set to 0. The constant δ is chosen so that the condition number (the ratio of the maximal and minimal singular values of a matrix) is p . The shape parameter β is generated in two steps. First, set each entry of β as either 0 or 1 according to the Bernoulli distribution with success probability 0.3. Next, replace each nonzero entry of β with a number randomly chosen from $\{3, 4, 5\}$. The parameter ξ is generated in the same way as Models I and II. To accommodate the sparsity in ξ , we update $\hat{\xi}$ by maximizing the penalized likelihood with ℓ_1 penalty, and Θ and β fixed at their sparse estimates. We used the same six (n, p) scenarios as in Models I and II. The averaged ROC curves for the four estimators based on 50 replicates are presented in Fig. 3. The figure shows that, in all cases, the skewed Gaussian graphical models and its refinement significantly outperform the Gaussian and the Gaussian copula graphical models. In addition, we compared means and standard errors (SE) of areas under curves (AUC) for Models I, II and III in the Table 1.

7. Application to breast cancer MicroRNA networks

In this section, we apply our skewed Gaussian graphical model to construct a MicroRNA expression network from the next generation sequencing data, and compare it with the Gaussian graphical model. MicroRNAs are a class of small noncoding RNAs that can control gene expressions. The abnormal expressions in MicroRNAs can be related to cancer. This data consists of counts of sequencing reads mapped back to a reference genome, so that the Gaussian assumption generally fails to hold.

We use the Level-III data obtained from the Cancer Genome Atlas (TCGA) portal (<http://tcga-data.nci.nih.gov/tcga/>). An open source software package, TCGA2STAT [35], is used to help obtain the TCGA data. This data consists of 842 subjects and 547 MicroRNAs, where the 842 subjects include 755 patients and 87 nonpatients. The data is preprocessed as follows:

- (i) 50% of the MicroRNAs that varied the least across the samples were filtered out, leaving 274 MicroRNAs in the sample.
- (ii) The F-ratio test is performed on patient data and nonpatient data to select the top 100 MicroRNAs that have the highest F-ratio.
- (iii) The samples are standardized by subtracting column means and divided by standard deviation for each column.

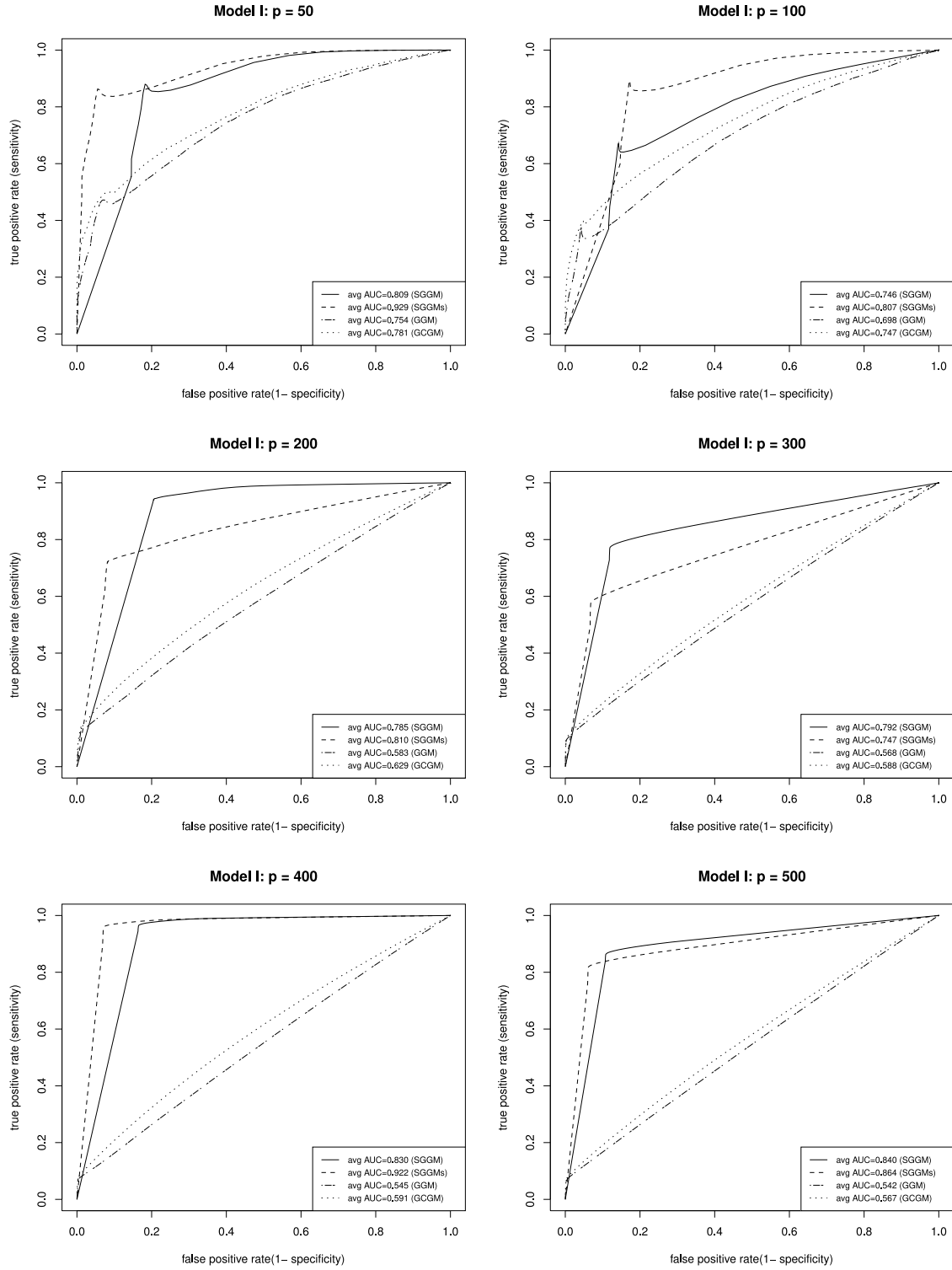


Fig. 1. Mean ROC curves for different estimators of Model I: the skewed Gaussian graphical model estimator (SSGM); refined skewed Gaussian graphical model with ℓ_1/ℓ_2 penalty estimator (SSGMs); the Gaussian graphical model (GGM); the Gaussian copula graphical model (GCGM).

We used two-step original estimator of Skewed Gaussian graphical model and the estimator of Gaussian graphical estimator from R package 'glasso' to construct two networks respectively. Tuning parameters for SSGM are selected

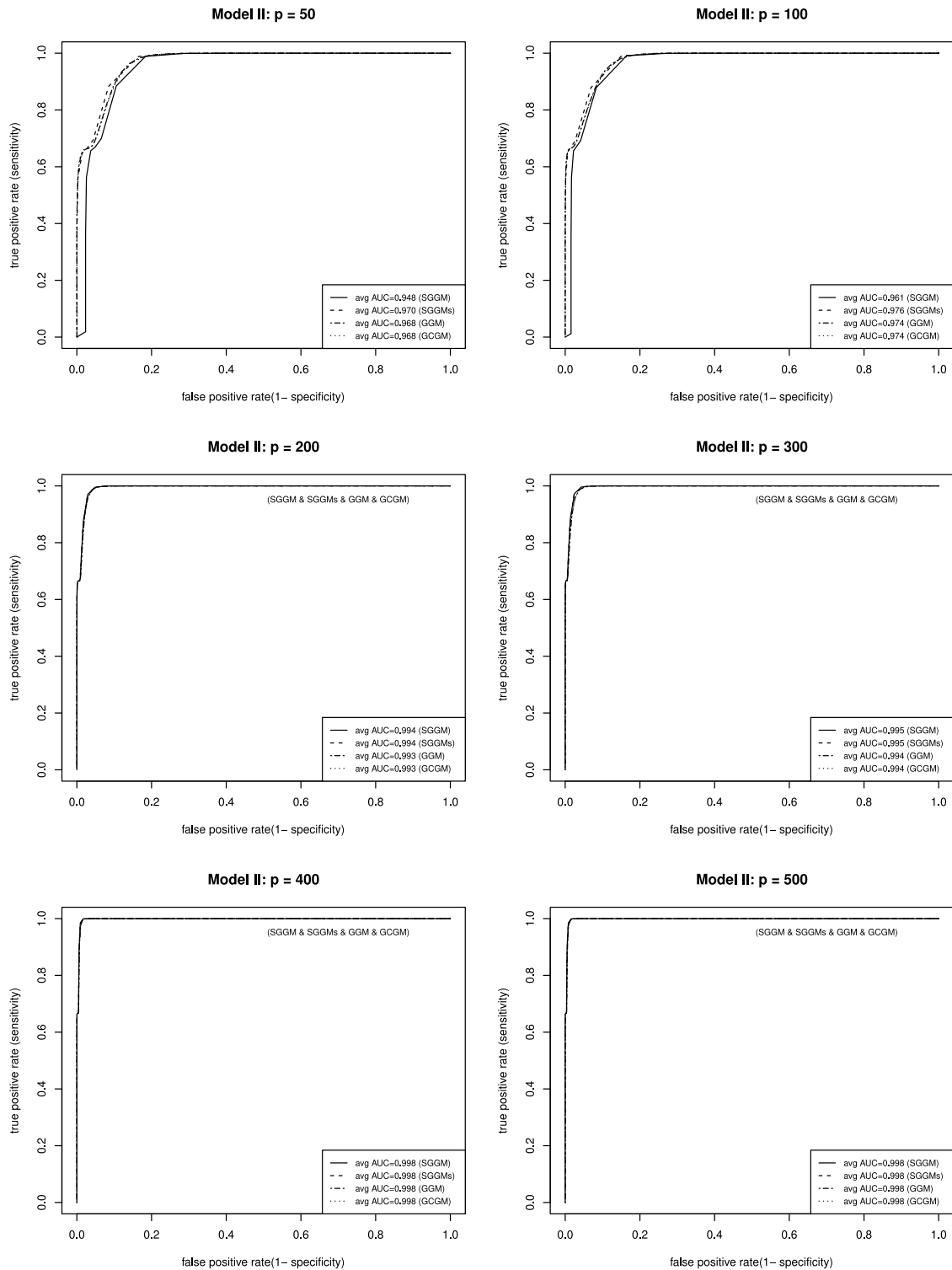


Fig. 2. Mean ROC curves for different estimators of Model II: the skewed Gaussian graphical model estimator (SGGM); refined skewed Gaussian graphical model with ℓ_1/ℓ_2 penalty estimator (SGGMs); the Gaussian graphical model (GGM); the Gaussian copula graphical model (GCGM). Since the curves for all four estimators are quite close, we indicated them by putting “SGGM & SGGMs & GGM & GCGM” beneath them.

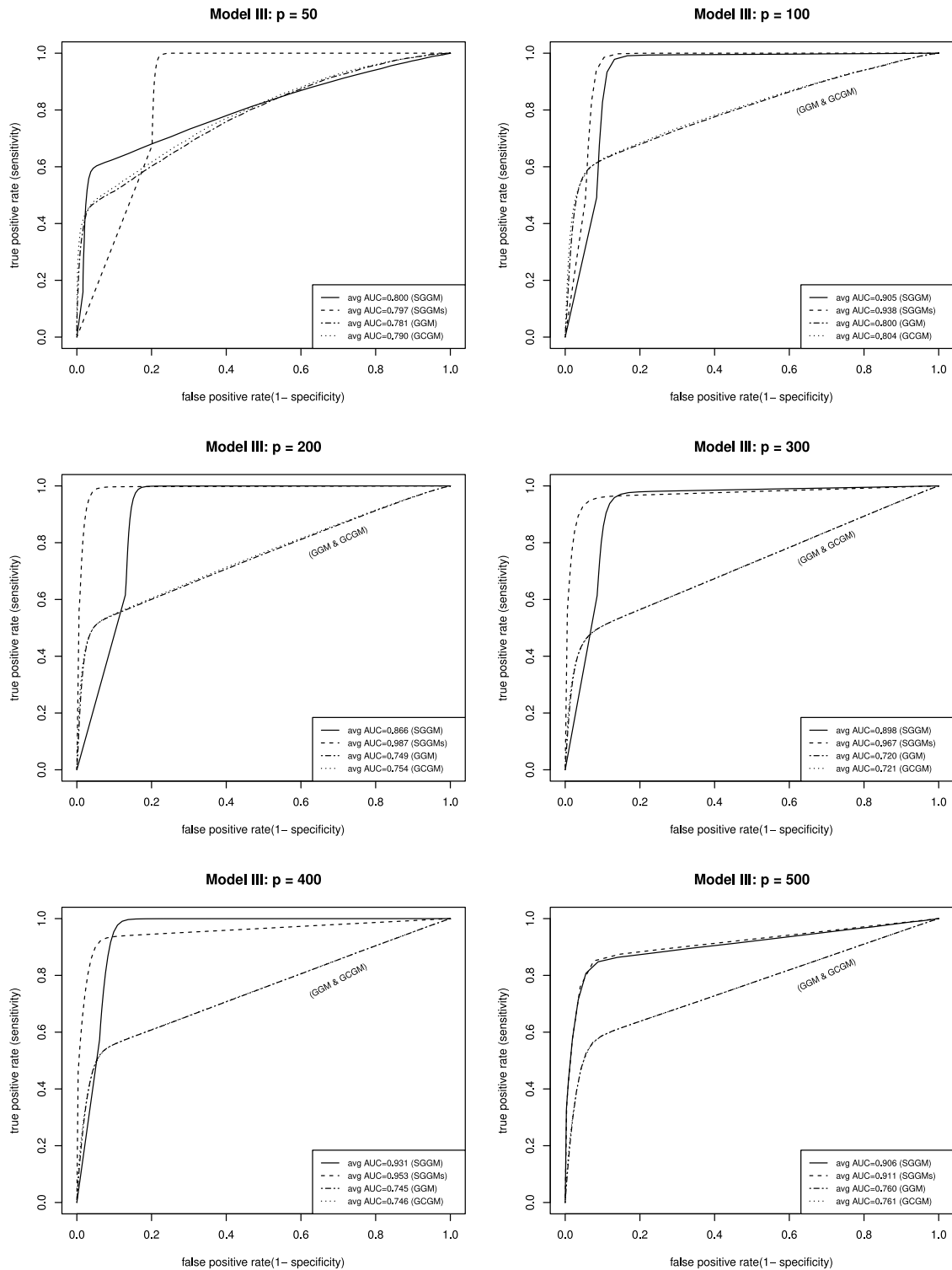


Fig. 3. Mean ROC curves for different estimators of Model III: the skewed Gaussian graphical model estimator (SGGM); refined skewed Gaussian graphical model with ℓ_1/ℓ_2 penalty estimator (SGGMs); the Gaussian graphical model (GGM); the Gaussian copula graphical model (GCGM). Since the curves for GGM and GCGM are quite close, we indicated them by putting "GGM & GCGM" beneath them.

Table 1

Comparison of average (SE) area under curve (AUC) over 50 replicates for three models based on four estimators: the skewed Gaussian graphical model estimator (SGGM); refined skewed Gaussian graphical model with ℓ_1/ℓ_2 penalty estimator (SGGMs); the Gaussian graphical model (GGM); the Gaussian copula graphical model (GCGM).

(p, n)	SGGM	SGGMs	GGM	GCGM
Model I				
(50, 1000)	0.809 (0.032)	0.929 (0.013)	0.754 (0.009)	0.781 (0.008)
(100, 1000)	0.746 (0.003)	0.807 (0.031)	0.698 (0.009)	0.747 (0.009)
(200, 2000)	0.785 (0.024)	0.810 (0.041)	0.583 (0.005)	0.629 (0.005)
(300, 2000)	0.792 (0.018)	0.747 (0.040)	0.568 (0.002)	0.588 (0.004)
(400, 3000)	0.830 (0.015)	0.922 (0.010)	0.545 (0.003)	0.591 (0.003)
(500, 3000)	0.840 (0.015)	0.864 (0.024)	0.542 (0.002)	0.567 (0.002)
Model II				
(50, 1000)	0.948 (0.015)	0.970 (0.003)	0.968 (0.002)	0.968 (0.002)
(100, 1000)	0.961 (0.008)	0.976 (0.002)	0.974 (0.002)	0.974 (0.002)
(200, 2000)	0.994 (0.000)	0.994 (0.000)	0.993 (0.000)	0.993 (0.000)
(300, 2000)	0.995 (0.000)	0.995 (0.000)	0.994 (0.000)	0.994 (0.000)
(400, 3000)	0.998 (0.000)	0.998 (0.000)	0.998 (0.000)	0.998 (0.000)
(500, 3000)	0.998 (0.000)	0.998 (0.000)	0.998 (0.000)	0.998 (0.000)
Model III				
(50, 1000)	0.800 (0.024)	0.797 (0.055)	0.781 (0.011)	0.790 (0.011)
(100, 1000)	0.905 (0.025)	0.938 (0.023)	0.800 (0.005)	0.804 (0.005)
(200, 2000)	0.866 (0.027)	0.987 (0.006)	0.749 (0.003)	0.754 (0.003)
(300, 2000)	0.898 (0.016)	0.967 (0.009)	0.720 (0.003)	0.721 (0.003)
(400, 3000)	0.931 (0.010)	0.953 (0.010)	0.745 (0.002)	0.746 (0.002)
(500, 3000)	0.906 (0.000)	0.911 (0.009)	0.760 (0.001)	0.761 (0.001)

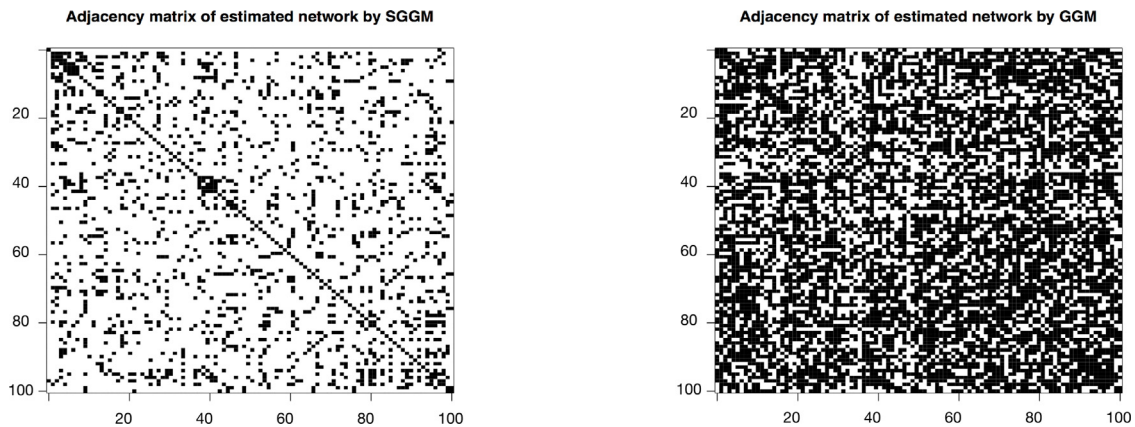


Fig. 4. (a) Adjacency matrix of the estimated network by two-step original estimator of SGGM; (b) Adjacency matrix of the estimated network by Gaussian graphical model using 'glasso' R package.

using BIC criterion and the tuning parameter for GGM is selected by cross validation under log-likelihood loss. Fig. 4 shows the adjacency matrices of the two networks for the patient group, with Fig. 4(a) corresponding to our method and Fig. 4(b) the Gaussian graphical model. The (i, j) th entry is dark if and only if the i th gene and j th gene are connected. We can see that Gaussian graphical model generates a much denser adjacency matrix than our method. In fact, the network estimated by the Gaussian graphical model is not sparse, with most of its nodes connected with each other. The estimated networks by our model and the Gaussian graphical model are shown in Fig. 5.

Since the true network is not available, one plausible way to compare the efficacy of the two methods is by investigating whether the hubs identified by a network have stronger relations with the important MicroRNAs selected by the F-ratio than the nonhubs. Specifically, for the 842 samples of any particular MicroRNA, we calculate the F-ratio of the between- and within-sum of squares of the two groups. In Fig. 6, we plot the logarithm of the F-ratio of each MicroRNA versus its degree, again for the patient group. The left panel corresponds to our network and the right panel corresponds to the Gaussian graphical model. The left panel shows that the MicroRNAs with more connections tend to have higher F-ratios, whereas the right panel shows no clear pattern at all.

Many of the hub MicroRNAs identified by our skewed Gaussian graphical model, such as mir-1037-3p, mir-145, mir-125b and mir-10b, have been found to be associated with breast cancer progression and metastasis previously. For example, upregulated expression of mir-1037-3p was identified in breast cancer tissue in [18]. They concluded that mir-1307-3p is a oncogenic MicroRNA that contributes to breast cancer development and progression. It is reported that

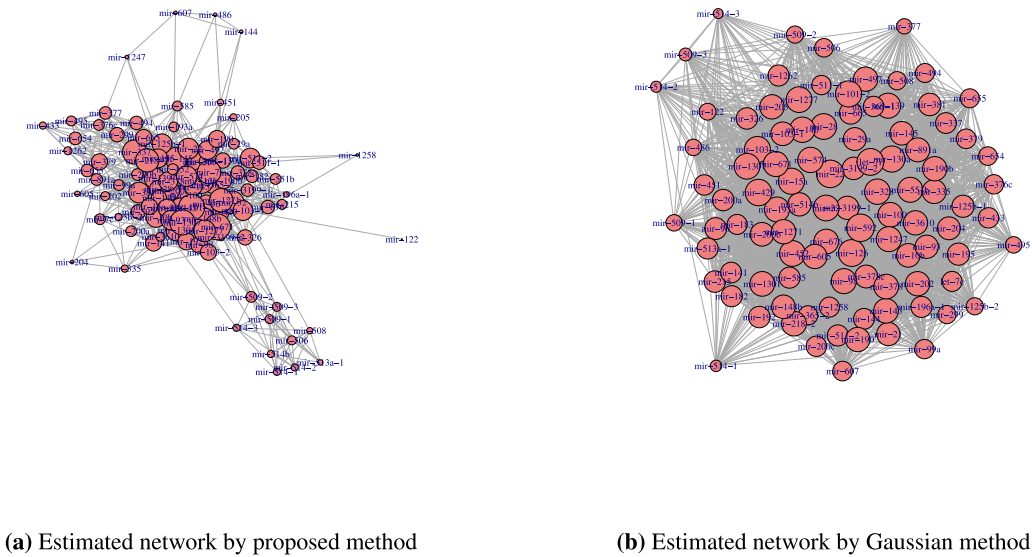


Fig. 5. (a) Breast cancer MicroRNA network estimated by skewed Gaussian graphical model (SGGM); (b) Breast cancer MicroRNA network estimated by Gaussian graphical model (GGM). The size of each node is proportion to the number of edges connected with other nodes.

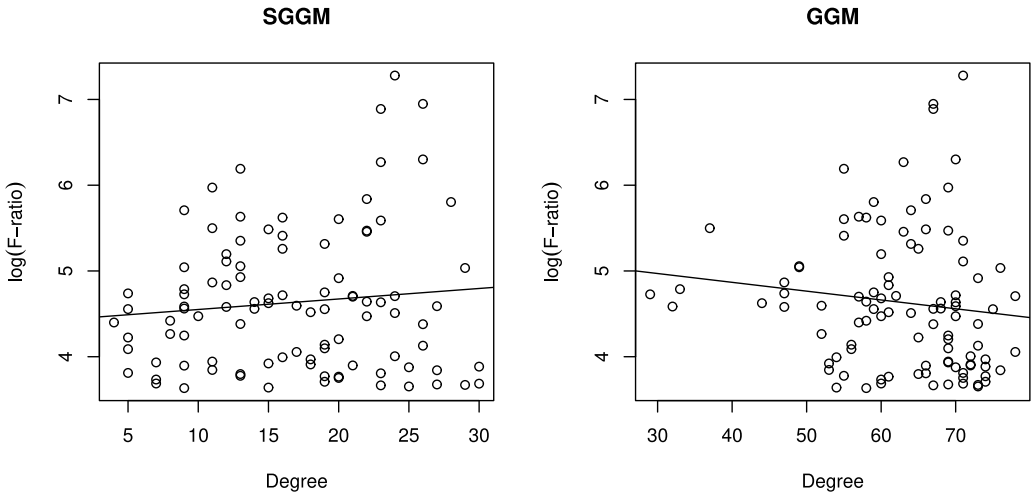


Fig. 6. $\log(F\text{-ratio})$ versus degree for each MicroRNA in the breast cancer MicroRNA network estimated by skewed Gaussian graphical model (SGGM) (left) and Gaussian graphical model (GGM) (right). The solid line is the regression line fitted by 100 top genes after removing one outlier.

the overall miRNA expressions could clearly separate normal tissues from cancer tissues, with the most significantly deregulated miRNAs being mir-125b, mir-145, mir-21, and mir-155 [21]. MicroRNA-10b (mir-10b) is found to be highly expressed in metastatic breast cancer cells and positively regulates cell migration and invasion [29]. Other hub microRNAs like mir-671, mir-1277 and mir-7i discovered by our model might be potential biomarkers which requires further investigation in the future and can provide certain directions for researchers in relevant fields to analyze breast cancer risk.

CRedit authorship contribution statement

Tianhong Sheng: Conceptualization, Methodology, Formal analysis, Software, Data curation, Writing – original draft. **Bing Li:** Conceptualization, Validation, Supervision, Writing – review & editing. **Eftychia Solea:** Conceptualization, Resources, Writing – review & editing.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jmva.2022.105129>. The supplementary material contains sparse estimation of the parameter Θ and the estimation of the location parameter ξ .

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